

2025 AMC 10B Problem 6 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 6. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 6

Problem:

The line $y = \frac{1}{3}x + 1$ divides the square region defined by $0 \leq x \leq 2$ and $0 \leq y \leq 2$ into an upper and a lower region. The line $x = a$ divides the lower region into two regions of equal area. Then a can be written as $\sqrt{s} - t$, where s and t are positive integers. What is $s + t$?

Answer Choices:

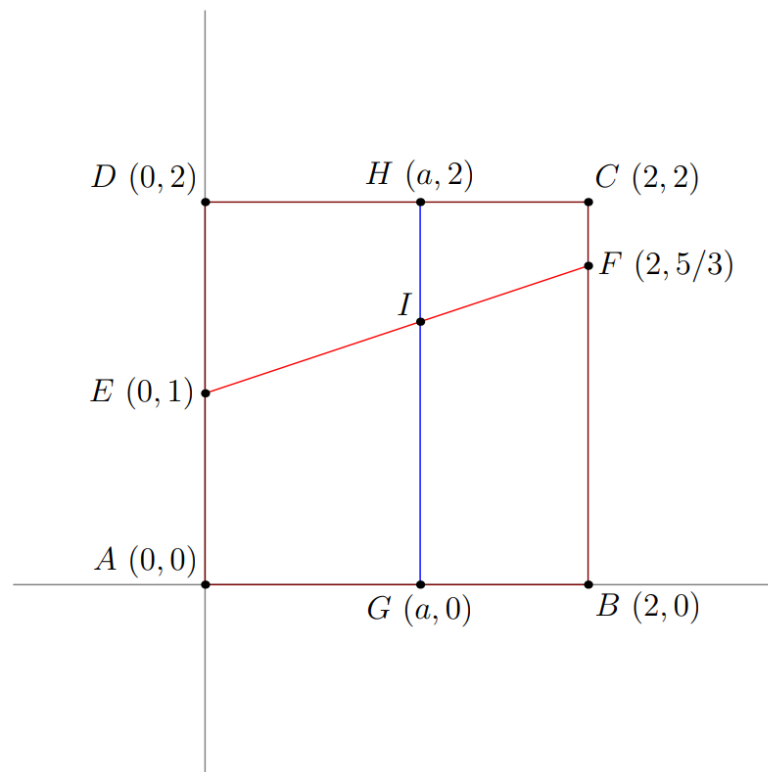
- A. 18
- B. 19
- C. 20
- D. 21
- E. 22

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Solution:



Take a look at the diagram above, and assign coordinates to the vertices as shown. Note that E is the point on $\ell : y = \frac{1}{3}x + 1$ when $x = 0$, or $(0, 1)$, and F is the point on ℓ when $x = 2$, or $\left(2, \frac{5}{3}\right)$. Note that the x -coordinate of I is a , so

$$I = \left(a, \frac{a}{3} + 1\right)$$

The blue line $x = a$ splits quadrilateral $AEFB$ in equal portions, so $[AEIG] = [GIFB]$. Note both of these are trapezoids - $AEIG$ has bases AE, GI and height AG , while $BGIF$ has bases BF, GI and height BG . Hence,

$$[AEIG] = \frac{1}{2}(AE + GI) \cdot AG = \frac{1}{2} \left(1 + 1 + \frac{a}{3}\right) \cdot a = a + \frac{a^2}{6}$$

$$[GIFB] = \frac{1}{2}(GI + BF) \cdot BG = \frac{1}{2} \left(\frac{5}{3} + 1 + \frac{a}{3} \right) \cdot (2 - a) = \frac{1}{6}(8 + a)(2 - a) = -\frac{a^2}{6} - a + \frac{8}{3}$$

We can therefore equate these to get

$$a + \frac{a^2}{6} = -\frac{a^2}{6} - a + \frac{8}{3} \quad a^2 + 6a - 8 = 0$$

This gives us $a = -3 \pm \sqrt{17}$ by the quadratic formula, but as $-3 - \sqrt{17} < 0$, $a = \sqrt{17} - 3$. Thus, $s + t = \boxed{\text{(C) } 20}$.

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